

THE WINDOW STATISTIC FOR THE ERDŐS–STRAUS EQUATION: EXACT ORDERS FOR BOUNDED WINDOWS AND THE ROLE OF PARITY

ANONYMOUS

ABSTRACT. For a prime $p \equiv 1 \pmod{4}$ and a positive integer $a \equiv 3 \pmod{4}$ put $x_a = (p+a)/4$. The equation $4/p = 1/x + 1/y + 1/z$ has a solution with denominator $x = x_a$ exactly when -1 or $-p$ is a ratio u/v modulo a of coprime u, v with $uv \mid x_a$. The least such a , the *window statistic* $a_{\min}(p)$, is finite for every p if and only if the Erdős–Straus conjecture holds. We study how often $a_{\min}(p)$ is large. For every fixed Z the number of primes $p \leq N$ with $a_{\min}(p) > Z$ is $\ll_Z N/(\log N)^{1+J(Z)/2}$, where $J(Z)$ is the number of $a \leq Z$ with $a \equiv 3 \pmod{4}$; this is the exponent of the natural random model. The bound is attained for $Z = 3$: at least $\gg x/(\log x)^{3/2}$ primes $p \leq x$ in Mordell’s hard classes have $a_{\min}(p) \geq 7$ (modulo cited sieve theorems). On the Elliott–Halberstam conjecture it is also attained for $Z = 7$. The parity of the number of bad prime factors, a congruence condition on p , is a necessary input: we exhibit two explicit sets of primes with the same sieve data at every level allowed by Bombieri–Vinogradov, of which one has window 3 failing for $\gg x/(\log x)^{3/2}$ elements and the other for none. For two windows our method needs a level of distribution close to 1. A model computation suggesting that Type I information of level $1/2$ plus parity may not suffice for two windows is recorded as a remark, with its limited scope; it is not a statement about primes. Nothing here proves the Erdős–Straus conjecture.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers. It suffices to treat primes $n = p$, and Mordell’s polynomial identities [13] settle every prime outside the six classes

$$p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}.$$

We call these primes *Mordell-hard*. Each of the six classes is a square modulo 3, 5, 7 and 8, so Mordell-hard primes satisfy $p \equiv 1 \pmod{24}$ and $\left(\frac{p}{3}\right) = \left(\frac{p}{5}\right) = \left(\frac{p}{7}\right) = 1$.

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1.1. The window statistic. Fix a prime $p \equiv 1 \pmod{4}$. For an integer x coprime to an odd modulus a let

$$\begin{aligned} \text{Rat}_a(x) &= \left\{ \prod_{r|x} r^{f_r} \bmod a : |f_r| \leq v_r(x) \right\} \\ &= \{ uv^{-1} \bmod a : u, v \geq 1, \gcd(u, v) = 1, uv \mid x \} \subseteq (\mathbb{Z}/a\mathbb{Z})^\times \end{aligned}$$

(the product is over primes r ; the two descriptions agree by putting the positive exponents into u and the negative ones into v). For $a \equiv 3 \pmod{4}$, $a \geq 3$, put

$$x_a = \frac{p+a}{4},$$

an integer. We say that the *window a succeeds* if $p \nmid x_a$ and $\text{Rat}_a(x_a) \cap \{-1, -p\} \neq \emptyset$, and that it *fails* otherwise. (For $a < 3p$ we have $x_a < p$, so $p \nmid x_a$ automatically.) Define

$$a_{\min}(p) = \min\{a \equiv 3 \pmod{4} : \text{window } a \text{ succeeds}\} \in \{3, 7, 11, \dots\} \cup \{\infty\}.$$

Theorem 2.2 below shows that (1) is solvable for $n = p$ if and only if $a_{\min}(p) < \infty$; the window a is the denominator $x = x_a$ of a solution coprime to p , and the two targets $-1, -p$ correspond to the two classical solution types. Thus the Erdős–Straus conjecture is the statement that the search $a = 3, 7, 11, \dots$ always terminates, and $a_{\min}(p)$ is its length. Numerically $a_{\min}(p)$ is small: for every Mordell-hard $p < 10^8$ one has $a_{\min}(p) \leq 107$ (Remark 9.3).

1.2. Results. Write $J(Z) = \#\{a \leq Z : a \equiv 3 \pmod{4}\} = \lfloor (Z+1)/4 \rfloor$. In the natural random model (Remark 9.1) each window fails with probability of order $(\log p)^{-1/2}$, roughly independently, so one expects

$$\#\{p \leq N : a_{\min}(p) > Z\} \approx \pi(N) (\log N)^{-J(Z)/2}. \quad (2)$$

Theorem A (Theorem 4.2, Corollary 4.3; PROVED). *For every fixed finite set A of positive integers $a \equiv 3 \pmod{4}$,*

$$\begin{aligned} \#\{3 \max A < p \leq N : p \equiv 1 \pmod{24}, -1 \notin \text{Rat}_a(x_a) \text{ for all } a \in A\} \\ \ll_A \frac{N}{(\log N)^{1+|A|/2}}. \end{aligned}$$

In particular $\#\{p \leq N : p \equiv 1 \pmod{24}, a_{\min}(p) > Z\} \ll_Z N/(\log N)^{1+J(Z)/2}$ for every fixed $Z \geq 3$. A uniform version holds for $Z = o(\log \log N)$ (Theorem 4.4).

The key input is a *half-set lemma* (Lemma 3.1): if a window fails, all prime factors of x_a lie in one of boundedly many explicit sets of exactly half of the reduced classes modulo a . Theorem A is then a fixed-dimension upper-bound sieve.

Theorem B (Theorem 6.1; PROVED modulo cited sieve theorems).

$$\#\{p \leq x : p \equiv 1 \pmod{840}, (p+3)/4 \text{ has no prime factor } \equiv 2 \pmod{3}\} \gg \frac{x}{(\log x)^{3/2}},$$

and every prime counted is Mordell-hard with $a_{\min}(p) \geq 7$.

Theorem C (Theorem 7.1; CONDITIONAL on Elliott–Halberstam). *Assume the Elliott–Halberstam conjecture. Then $\gg x/(\log x)^2$ Mordell-hard primes $p \leq x$ have $a_{\min}(p) \geq 11$.*

Together with Theorem A these give the exact orders

$$\begin{aligned} \#\{p \leq x : p \equiv 1 \pmod{24}, a_{\min}(p) \geq 7\} &\asymp \frac{x}{(\log x)^{3/2}}, \\ \#\{p \leq x : p \equiv 1 \pmod{24}, a_{\min}(p) \geq 11\} &\asymp \frac{x}{(\log x)^2}, \end{aligned}$$

the first modulo the cited sieve theorems, the second (lower bound only) on Elliott–Halberstam (Corollary 7.2).

The proofs of Theorems B and C rest on a congruence fact (Lemma 2.5): the number of prime factors of x_a that are non-residues modulo a , counted with multiplicity, has parity $(\frac{p}{a})$. For one window and a sieve of dimension $1/2$ this removes the parity obstruction. The next result shows that this congruence input is not optional.

Proposition D (Proposition 8.2; PROVED). *Let $\mathcal{A}^\pm = \{p \leq x : p \equiv 1 \pmod{8}, p \equiv 1 \pmod{35}, (\frac{p}{3}) = \pm 1\}$ and $P_3 = \{\ell \text{ prime} : \ell \equiv 2 \pmod{3}\}$. The two sequences $\{(p+3)/4 : p \in \mathcal{A}^\pm\}$ have the same sieve data with respect to P_3 up to remainders admissible at level $x^{1/2}(\log x)^{-B}$ (Bombieri–Vinogradov), yet window 3 fails for no $p \in \mathcal{A}^-$ and for $\gg x/(\log x)^{3/2}$ primes $p \in \mathcal{A}^+$.*

Proposition D is a direct instance of Selberg’s parity example; it shows that parity is necessary, not that it is insufficient. Two windows (3 and 7) form a sieve problem of dimension one, for which our method needs a level of distribution close to 1; this is why our proof of Theorem C is conditional. We do not prove that unconditional methods must fail for two windows. The problem is a window analogue of the Friedlander–Iwaniec hyperbolic prime number theorem [3], whose lower bound is likewise conditional (Lemma 8.1, Remark 7.3). Section 9 records, with an explicit MODEL-ONLY label, a finite linear-programming computation in a coarse discretised model in which Type I information of level $1/2$ plus parity does not force any configuration with both windows clean.

1.3. Relation to known results. The exceptional set of the Erdős–Straus equation is known to be small on average: Vaughan [20] proved $E(N) \ll N \exp(-c(\log N)^{2/3})$ by a large-sieve argument over many moduli, which is the classical form of the window sieve of Section 4. Such bounds concern $\{a_{\min}(p) = \infty\}$. This note concerns the much larger sets $\{a_{\min}(p) > Z\}$ for

fixed or slowly growing Z ; the comparison is made precise in Remark 4.5. The lower bounds (Theorems B, C) follow the half-dimensional sieve method of Iwaniec [7] and Friedlander–Iwaniec [3]; we claim no novelty of sieve technique.

1.4. Status and conventions. Labels: PROVED means a complete proof is given here, possibly using cited theorems that are listed explicitly; CONDITIONAL names the hypothesis; EVIDENCE is numerical; MODEL statements concern an explicit heuristic model and not the primes. All cited external theorems are collected in Section 5, each with a note on whether its exact statement was checked against a source. The arguments have had internal checks only; they have not been externally refereed. The Erdős–Straus conjecture is not proved here, and none of the results bounds $a_{\min}(p)$ for every p .

Throughout, p denotes a prime, r, ℓ denote primes, $(\frac{\cdot}{a})$ is the Jacobi symbol, $\Omega(n)$ counts prime factors with multiplicity, and implied constants depend only on the indicated parameters.

2. THE WINDOW FRAME

2.1. The window criterion.

Lemma 2.1 (two unit fractions). *Let r, s be coprime positive integers. Then $r/s = 1/y + 1/z$ has a solution in positive integers if and only if there are coprime positive integers d_1, d_2 with $d_1 d_2 \mid s$ and $r \mid d_1 + d_2$; the solution is then $y = s(d_1 + d_2)/(rd_1)$, $z = s(d_1 + d_2)/(rd_2)$.*

Proof. Sufficiency is the identity $1/y + 1/z = r(d_1 + d_2)/(s(d_1 + d_2))$; both denominators are integers because $d_1 d_2 \mid s$, $\gcd(d_1, d_2) = 1$ and $r \mid d_1 + d_2$, $\gcd(r, d_i) \mid \gcd(r, s) = 1$. Conversely, write $y = gy_1$, $z = gz_1$ with $\gcd(y_1, z_1) = 1$. Then $rgy_1 z_1 = s(y_1 + z_1)$. As $\gcd(y_1 + z_1, y_1 z_1) = 1$, we get $y_1 z_1 \mid s$; and as $\gcd(r, s) = 1$, we get $r \mid y_1 + z_1$. Take $d_1 = z_1$, $d_2 = y_1$. $\square$

Theorem 2.2 (window criterion; PROVED). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \equiv 3 \pmod{4}$ be positive, put $x = x_a = (p + a)/4$ and suppose $p \nmid x$. Then $\gcd(a, x) = 1$, and:*

- (1) *If $u, v \geq 1$ are coprime, $uv \mid x$ and $u \equiv -v \pmod{a}$, then*

$$\frac{4}{p} = \frac{1}{x} + \frac{au}{px(u+v)} + \frac{av}{px(u+v)} \quad (3)$$

is a solution of (1) in positive integers.

- (2) *If $u, v \geq 1$ are coprime, $uv \mid x$ and $u \equiv -pv \pmod{a}$, then*

$$\frac{4}{p} = \frac{1}{x} + \frac{au}{px(u+pv)} + \frac{av}{x(u+pv)} \quad (4)$$

is a solution of (1) in positive integers.

- (3) *Conversely, if $4/p = 1/x + 1/y + 1/z$ with $p \nmid x$, then $x = x_a$ for $a = 4x - p$, which is a positive integer $\equiv 3 \pmod{4}$, and $\text{Rat}_a(x) \cap \{-1, -p\} \neq \emptyset$.*

Consequently (1) is solvable for $n = p$ if and only if $a_{\min}(p) < \infty$.

Proof. Since $4x - a = p$, $\gcd(a, x) = \gcd(p, x) = 1$. Also a is odd and coprime to p , so $4/p - 1/x = a/(px)$ is in lowest terms. By Lemma 2.1 with $(r, s) = (a, px)$, $a/(px) = 1/y + 1/z$ is solvable if and only if there are coprime d_1, d_2 with $d_1 d_2 \mid px$ and $a \mid d_1 + d_2$. Taking $(d_1, d_2) = (u, v)$ gives (3), and $(d_1, d_2) = (u, pv)$ gives (4); in both cases the denominators $px(d_1 + d_2)/(ad_i)$ are integers by Lemma 2.1, and one checks that the two displayed fractions are $1/y, 1/z$. This proves (1) and (2).

For (3): $1/x < 4/p$ gives $a = 4x - p > 0$, and $a \equiv -p \equiv 3 \pmod{4}$. Lemma 2.1 gives coprime d_1, d_2 with $d_1 d_2 \mid px$ and $a \mid d_1 + d_2$. Since $\gcd(d_1, d_2) = 1$ and $p \nmid px$, the prime p divides at most one of them. If $p \nmid d_1 d_2$, then $d_1 d_2 \mid x$ and $d_1 \equiv -d_2 \pmod{a}$, so $-1 \in \text{Rat}_a(x)$. If $p \mid d_2$, say $d_2 = pv$, then $d_1 v \mid x$, $\gcd(d_1, v) = 1$ and $d_1 \equiv -pv$, so $d_1 v^{-1} \equiv -p$ and $-p \in \text{Rat}_a(x)$; the case $p \mid d_1$ is the same with d_1, d_2 exchanged.

Finally, if (1) holds for $n = p$, not all of x, y, z are divisible by p : otherwise multiplying by p would write 4 as a sum of three unit fractions. Relabel so that $p \nmid x$ and apply (3); then window $a = 4x - p$ succeeds. The converse is (1)–(2). $\square$

Solutions from (3) have two denominators divisible by p and those from (4) exactly one; these are the Type II and Type I solutions of the literature [2]. The criterion is a reformulation of the classical parametrisations of these two types (compare [2, §2]); we use it only in the form of Theorem 2.2. We do not use the targets $-p$ in any upper bound below: all upper bounds concern the larger event that $-1 \notin \text{Rat}_a(x_a)$.

2.2. Window reciprocity and the parity of the bad part.

Lemma 2.3 (window reciprocity; PROVED). *Let $p \equiv 1 \pmod{8}$ be prime, $a \equiv 3 \pmod{4}$ with $0 < a < 3p$, and $x = x_a$. Every prime $r \mid x$ is coprime to a and satisfies $\left(\frac{r}{a}\right) = \left(\frac{r}{p}\right)$.*

Proof. If $r \mid a$ then $r \mid 4x - a = p$, so $r = p$; but $r \leq x < p$. If r is odd, Jacobi reciprocity with $(a-1)/2$ odd gives $\left(\frac{r}{a}\right) = \left(\frac{-1}{r}\right) \left(\frac{a}{r}\right) = \left(\frac{-a}{r}\right)$; since $r \mid p + a$ this is $\left(\frac{p}{r}\right)$, which equals $\left(\frac{r}{p}\right)$ as $p \equiv 1 \pmod{4}$. If $r = 2$ then $8 \mid p + a$, so $a \equiv 7 \pmod{8}$ and $\left(\frac{2}{a}\right) = 1 = \left(\frac{2}{p}\right)$. $\square$

Call a prime $r \nmid a$ *a-bad* if $\left(\frac{r}{a}\right) = -1$ and *a-good* otherwise, and let $\Omega_a^-(n)$ be the number of *a-bad* prime factors of n counted with multiplicity. Since $\left(\frac{r}{a}\right)$ depends only on r modulo a , and $\left(\frac{\cdot}{a}\right)$ is a non-principal character of $(\mathbb{Z}/a\mathbb{Z})^\times$ ($a \equiv 3 \pmod{4}$ is not a square), the set of *a-bad* primes is a union of half of the reduced classes modulo a , of relative density $1/2$, and does not depend on p .

Lemma 2.4 (clean windows fail; PROVED). *Let $p \equiv 1 \pmod{8}$, $a \equiv 3 \pmod{4}$, $0 < a < 3p$. If x_a has no *a-bad* prime factor, window a fails.*

Proof. $u \mapsto \left(\frac{u}{a}\right)$ is a character of $(\mathbb{Z}/a\mathbb{Z})^\times$, so every element of $\text{Rat}_a(x_a)$ has symbol $+1$. But $\left(\frac{-1}{a}\right) = -1$ as $a \equiv 3 \pmod{4}$, and $\left(\frac{-p}{a}\right) = -\left(\frac{4x_a}{a}\right) = -\prod_{r|x_a} \left(\frac{r}{a}\right)^{v_r(x_a)} = -1$. $\square$

By Lemma 2.3, “ x_a has no a -bad prime factor” is the same as “every prime factor of x_a is a quadratic residue modulo p ”. We call such a window *clean*. Writing the condition through $\left(\frac{\cdot}{a}\right)$ rather than $\left(\frac{\cdot}{p}\right)$ makes it a sifting condition on the shifted prime x_a by a fixed set of primes of density $1/2$.

Lemma 2.5 (parity of the bad part; PROVED). *If $\gcd(x_a, a) = 1$ then $(-1)^{\Omega_a^-(x_a)} = \left(\frac{x_a}{a}\right) = \left(\frac{p}{a}\right)$.*

Proof. Multiplicativity gives $\left(\frac{x_a}{a}\right) = \prod_r \left(\frac{r}{a}\right)^{v_r(x_a)} = (-1)^{\Omega_a^-(x_a)}$, and $4x_a \equiv p \pmod{a}$ with $\left(\frac{4}{a}\right) = 1$. $\square$

Lemma 2.6 (window 3). *Let $p \equiv 1 \pmod{24}$. Window 3 fails if and only if $x_3 = (p+3)/4$ has no prime factor $\equiv 2 \pmod{3}$. Hence $a_{\min}(p) \geq 7$ exactly in this case.*

Proof. Modulo 3 both targets equal 2 ($-p \equiv -1$). The prime 3 does not divide x_3 as $p \equiv 1 \pmod{3}$. $\text{Rat}_3(x_3)$ is generated by the classes of the prime factors of x_3 , so it contains 2 if and only if some prime factor is $\equiv 2 \pmod{3}$. (The “if” direction is also Lemma 2.4.) $\square$

3. THE HALF-SET LEMMA

Let $a \equiv 3 \pmod{4}$, $a \geq 3$, and $G = (\mathbb{Z}/a\mathbb{Z})^\times$. Since some prime factor of a is $\equiv 3 \pmod{4}$, -1 is not a square in G ; and $g \neq -g$ as a is odd. The Klein four-group $\{g \mapsto g, g \mapsto g^{-1}, g \mapsto -g, g \mapsto -g^{-1}\}$ acts on G , and its orbits are of two kinds:

- 4-orbits $\{g, g^{-1}, -g, -g^{-1}\}$ when $g^2 \neq 1$ (the four elements are distinct because $g^2 \neq \pm 1$);
- 2-orbits $\{g, -g\}$ when $g^2 = 1$; one of them is $\{1, -1\}$.

Split every 4-orbit as $\{g, g^{-1}\} \sqcup \{-g, -g^{-1}\}$ and every 2-orbit as $\{g\} \sqcup \{-g\}$. A *selection* σ chooses one half of every orbit, subject to choosing $\{1\}$ from $\{1, -1\}$; let $S_\sigma \subseteq G$ be the union of the chosen halves. Since a is odd, $g^2 = 1$ has exactly $2^{\omega(a)}$ solutions, so there are $(\varphi(a) - 2^{\omega(a)})/4$ orbits of size four and $2^{\omega(a)-1}$ of size two. Hence

$$\begin{aligned} |S_\sigma| &= \frac{1}{2}\varphi(a) \text{ for every } \sigma, & \#\{\sigma\} &= 2^{\beta(a)}, \\ \beta(a) &= \frac{\varphi(a) - 2^{\omega(a)}}{4} + 2^{\omega(a)-1} - 1 \leq \frac{\varphi(a)}{4} + 2^{\omega(a)-2}. \end{aligned} \tag{5}$$

For x coprime to a let $C(x) \subseteq G$ be the set of classes of the prime factors of x .

Lemma 3.1 (half-set lemma; PROVED). *Let $a \equiv 3 \pmod{4}$, $a \geq 3$, and $\gcd(x, a) = 1$. If $-1 \notin \text{Rat}_a(x)$, then $C(x) \subseteq S_\sigma$ for some selection σ . In particular this holds whenever window a fails and $p \nmid x_a$, with $x = x_a$.*

Proof. For distinct primes $r, s \mid x$ the set $\text{Rat}_a(x)$ contains r , rs and rs^{-1} (take $(u, v) = (r, 1), (rs, 1), (r, s)$). Let $g, h \in C(x)$, possibly equal. If g, h are the classes of distinct primes, then $-1 \notin \text{Rat}_a(x)$ gives $gh \neq -1$ and $gh^{-1} \neq -1$. If $g = h$, then $gh = g^2 \neq -1$ because -1 is not a square, and $gh^{-1} = 1 \neq -1$. Also $-1 \notin C(x)$. Hence

$$C(x) \cap (-C(x)) = \emptyset, \quad C(x) \cap (-C(x)^{-1}) = \emptyset, \quad -1 \notin C(x).$$

In a 4-orbit, an element $g \in C(x)$ excludes $-g$ and $-g^{-1}$ from $C(x)$, and so does $g^{-1} \in C(x)$; so $C(x)$ meets the orbit inside one half. In a 2-orbit $\{g, -g\}$ the two elements exclude each other, and in $\{1, -1\}$ the element -1 is excluded. Choose σ accordingly, arbitrarily on orbits that $C(x)$ misses. $\square$

Remark 3.2. (i) For prime a the selection with S_σ equal to the squares is the clean case of Lemma 2.4. Failures need not be clean: for $a = 7$, $x = 17$, $p = 4 \cdot 17 - 7 = 61$ one has $C(x) = \{3\}$ and $\text{Rat}_7(17) = \{1, 3, 5\}$, which contains neither $-1 \equiv 6$ nor $-p \equiv 2$, although $\left(\frac{17}{7}\right) = -1$. Such a failure is confined to the non-subgroup half-set $\{1, 3, 5\}$. (ii) The lemma has no exceptions coming from exponent multiplicities and holds for composite a . Every failure is a confinement of all prime factors to a set of exactly half of the classes; this is what makes the exponent in Theorem 4.2 exact. (iii) Only the target -1 and products of at most two primes were used.

4. UPPER BOUNDS: FIXED-SET STACKING

4.1. A large-sieve lemma.

Lemma 4.1. *Let I be an interval containing $X \geq 2$ integers, $\mathcal{P}$ a finite set of primes, and for $\ell \in \mathcal{P}$ let Ω_ℓ be a set of $\nu(\ell) < \ell$ residue classes modulo ℓ . Let $S \subseteq I$ avoid Ω_ℓ modulo ℓ for every $\ell \in \mathcal{P}$. Put $V = \prod_{\ell \in \mathcal{P}} (1 - \nu(\ell)/\ell)$ and $\Lambda = \sum_{\ell \in \mathcal{P}} \nu(\ell) \log \ell / \ell$. If $\Lambda \leq (\log X)/4$, then $|S| \leq 4XV$.*

Proof. By the arithmetic large sieve (Cited Theorem 5.1), for every $Q \geq 1$,

$$|S| \leq \frac{X + Q^2}{H(Q)}, \quad H(Q) = \sum_{q \leq Q} g(q), \quad g(q) = \mu^2(q) \prod_{\ell \mid q} \frac{\nu(\ell)}{\ell - \nu(\ell)},$$

the sum being over q composed of primes of $\mathcal{P}$. The full sum is $\sum_q g(q) = \prod_{\ell \in \mathcal{P}} (1 + \nu(\ell)/(\ell - \nu(\ell))) = V^{-1}$. The weights $Vg(q)$ are the law of the random squarefree number $q = \prod_{\ell \in \mathcal{P}} \xi_\ell \ell$, where the ξ_ℓ are independent Bernoulli variables with $\Pr(\xi_\ell = 1) = \nu(\ell)/\ell$. Then $\mathbb{E} \log q = \Lambda$, and Markov's inequality gives $\Pr(\log q > \log Q) \leq \Lambda / \log Q \leq \frac{1}{2}$ for $Q = X^{1/2}$. Hence $H(X^{1/2}) \geq V^{-1}/2$ and $|S| \leq 2X \cdot 2V$. $\square$

4.2. The stacking theorem.

Theorem 4.2 (fixed-set stacking; PROVED). *Let A be a fixed finite set of positive integers $a \equiv 3 \pmod{4}$, and $J = |A|$. Then (the restriction $p > 3 \max A$ ensures $\gcd(x_a, a) = 1$, so that $\text{Rat}_a(x_a)$ is defined)*

$$\#\{3 \max A < p \leq N : p \equiv 1 \pmod{24}, -1 \notin \text{Rat}_a(x_a) \ \forall a \in A\} \ll_A \frac{N}{(\log N)^{1+J/2}}.$$

Proof. It suffices to count $p \in (N/2, N]$ with N large and sum over dyadic ranges. For $p > 3 \max A$ we have $x_a < p$ and $\gcd(x_a, a) = \gcd(p, a) = 1$, so Lemma 3.1 applies: each counted p has a selection vector $\sigma = (\sigma_a)_{a \in A}$ with $C(x_a) \subseteq S_{\sigma_a}$ for every $a \in A$. There are $\prod_{a \in A} 2^{\beta(a)} = O_A(1)$ vectors; fix one.

Write $p = 24t + 1$; then t runs over an interval I with $X \asymp N$ integers. Put $y = 2 \max A + 24$, $\log z = (\log X)/(5(J+1))$, $\mathcal{P} = \{\ell \text{ prime} : y < \ell \leq z\}$, and let Ω_ℓ consist of the class $24t + 1 \equiv 0 \pmod{\ell}$ and, for every $a \in A$ with $(\ell \bmod a) \notin S_{\sigma_a}$, the class $24t + 1 + a \equiv 0 \pmod{\ell}$. Every counted t avoids Ω_ℓ : $\ell \leq z < N/2 < p$ gives $\ell \nmid p$; and if $\ell \mid p + a$ then, ℓ being odd, $\ell \mid x_a$, so $(\ell \bmod a) \in C(x_a) \subseteq S_{\sigma_a}$. The listed classes are distinct, since $\ell > y$ exceeds every a and every $|a - a'|$, and $\ell \nmid 24$. Thus $\nu(\ell) = 1 + \#\{a \in A : (\ell \bmod a) \notin S_{\sigma_a}\} \leq J + 1 < \ell$, and $\Lambda \leq (J+1) \sum_{\ell \leq z} \log \ell / \ell \leq (J+1)(\log z + O(1)) \leq (\log X)/4$ for large N .

Each $G \setminus S_{\sigma_a}$ is a union of $\varphi(a)/2$ reduced classes modulo a , so Mertens' theorem in arithmetic progressions to the fixed moduli $a \in A$ (Cited Theorem 5.2) gives

$$\sum_{\ell \in \mathcal{P}} \frac{\nu(\ell)}{\ell} = \left(1 + \frac{J}{2}\right) \log \log z + O_A(1) = \left(1 + \frac{J}{2}\right) \log \log N + O_A(1).$$

By Lemma 4.1, $|S| \leq 4XV \leq 4X \exp(-\sum_{\ell \in \mathcal{P}} \nu(\ell)/\ell) \ll_A N(\log N)^{-1-J/2}$. Summing over the $O_A(1)$ vectors σ and over dyadic ranges (primes $p \leq \sqrt{N}$ contribute trivially) proves the theorem. $\square$

Corollary 4.3 (tail of $a_{\min}$; PROVED). *For every fixed $Z \geq 3$,*

$$\begin{aligned} \#\{p \leq N : p \equiv 1 \pmod{24}, a_{\min}(p) > Z\} &\ll_Z \frac{N}{(\log N)^{1+J(Z)/2}} \\ &= \pi(N)(\log N)^{-J(Z)/2} \cdot O_Z(1). \end{aligned}$$

Proof. For $p > 3Z$ every window $a \leq Z$ has $p \nmid x_a$; $a_{\min}(p) > Z$ means that all of them fail, in particular $-1 \notin \text{Rat}_a(x_a)$. Apply Theorem 4.2 with $A = \{a \leq Z : a \equiv 3 \pmod{4}\}$. $\square$

For example, the number of $p \leq N$, $p \equiv 1 \pmod{24}$, with $a_{\min}(p) > 7$, > 11 , > 23 is $\ll N/(\log N)^2$, $\ll N/(\log N)^{5/2}$, $\ll N/(\log N)^4$ respectively. Running the same argument with $p = 4t + 1$ covers all $p \equiv 1 \pmod{4}$; we restrict to $p \equiv 1 \pmod{24}$ because the other primes are settled by Mordell's identities.

4.3. Uniformity in Z . The union over selection vectors costs $2^{\beta_{\text{tot}}(Z)}$ with $\beta_{\text{tot}}(Z) = \sum_{a \leq Z, a \equiv 3(4)} \beta(a) = O(Z^2)$, which is affordable while $Z \ll \log \log N$.

Theorem 4.4 (uniform stacking; PROVED modulo Siegel–Walfisz). *Let $L = \log \log N$, $\mathcal{L} = \log N$, and fix $C_0 \geq 1$. Uniformly for $3 \leq Z \leq C_0 L$ and $N \geq N_0(C_0)$, with $J = J(Z)$,*

$$\begin{aligned} & \#\{N/2 < p \leq N : p \equiv 1 \pmod{24}, a_{\min}(p) > Z\} \\ & \leq \frac{N}{\mathcal{L}} \exp\left(-\frac{J}{2}(L - 3 \log L - C) + \beta_{\text{tot}}(Z) \log 2 + O(\log L)\right), \end{aligned}$$

where C and the implied constant depend only on C_0 . In particular the left side is $\pi(N)(\log N)^{-(1/2-o(1))J(Z)}$ uniformly for $Z = o(L)$.

Proof. Repeat the proof of Theorem 4.2 with $y = \exp(L^2)$ and $\log z = \mathcal{L}/(5(J+1))$. Now $y > 2Z+24$, so the classes are distinct and $\nu(\ell) \leq J+1 < \ell$; and $\Lambda \leq (J+1)(\log z + O(1)) \leq \mathcal{L}/4$ because $J \leq C_0 L$. The moduli satisfy $a \leq C_0 L = C_0(\log y)^{1/2}$, so the Siegel–Walfisz theorem (Cited Theorem 5.3) and partial summation give

$$\sum_{\substack{y < \ell \leq z \\ \ell \equiv c \pmod{a}}} \frac{1}{\ell} = \frac{1}{\varphi(a)} \log \frac{\log z}{\log y} + O(e^{-c_1 L})$$

uniformly in $a \leq Z$ and $\gcd(c, a) = 1$, with $c_1 = c_1(C_0) > 0$. Summing over at most $\sum_{a \leq Z} \varphi(a)/2 = O(L^2)$ classes, the total error is $o(1)$, and $\sum_{\ell \in \mathcal{P}} \nu(\ell)/\ell = (1 + J/2) \log(\log z / \log y) + o(1)$. Here $\log(\log z / \log y) = L - 2 \log L - \log(5(J+1)) \geq L - 3 \log L - C$. Lemma 4.1 bounds the count for one selection vector by $4X \exp(-(1 + J/2)(L - 3 \log L - C) + o(1))$, and $e^{-(L-3 \log L-C)} = e^C L^3 / \mathcal{L}$. Summing over the $2^{\beta_{\text{tot}}(Z)}$ selection vectors gives the bound. For the last claim note $\beta_{\text{tot}}(Z) = O(Z^2) = o(JL)$ when $Z = o(L)$. $\square$

Remark 4.5 (comparison with the literature). Since $\sum_{a \leq Z, a \equiv 3(4)} \varphi(a) = (1 + o(1))Z^2/\pi^2$, one has $\beta_{\text{tot}}(Z) = (1 + o(1))Z^2/(4\pi^2)$, and optimising the exponent $-(Z/8)L + (\log 2/4\pi^2)Z^2$ at $Z \approx \pi^2 L/(4 \log 2) \approx 3.56L$ (inside the range with $C_0 = 4$) gives $\#\{p \leq N : p \equiv 1 \pmod{24}, a_{\min}(p) > Z\} \leq N \exp(-(\pi^2/(64 \log 2) - o(1))(\log \log N)^2)$.

The mechanism of this section — sieving the shifted primes $p + a$ by residue conditions modulo many windows a at once — is classical. Vaughan [20] used the large sieve in this way to prove that the number $E(N)$ of $n \leq N$ for which (1) is unsolvable satisfies $E(N) \ll N \exp(-c(\log N)^{2/3})$; see [2] for counts of solutions. (Vaughan’s paper could not be consulted in this project; we use only its published bound.) The set $\{p : a_{\min}(p) = \infty\}$ of counterexamples is therefore far sparser than anything proved here, and an unpublished argument in this project’s notes obtains the shape $N \exp(-c(\log \log N)^2)$ for that set from windows up to $\delta \log N$ [15, Thm. 12.2],

and tails $N \exp(-c(\log N)^\theta \log \log N)$ for the failure of all windows up to $(\log N)^\theta$, $\theta < \log 3/(1 + \log 3)$ [15, Thms. 14.4, 14.9]; the review status of these notes sections is not recorded, and we do not reproduce them. The sets $\{a_{\min}(p) > Z\}$ studied here, with Z fixed or slowly growing, are much larger, and what is new, as far as we know, is the exact exponent $1 + J(Z)/2$ in Theorem 4.2, together with its sharpness for $Z = 3$ (and $Z = 7$ on EH). The ingredient that makes the exponent exact is the half-set lemma, which removes the exponent-budget exceptions that limited earlier window majorants in this project to a single excluded class per window [15, §71]. None of these bounds approaches $a_{\min}(p) < \infty$ for all p .

5. CITED RESULTS

We list every external result used, with a note on what was checked. “Checked” means that the exact statement was compared with an archived copy of the source; “from memory” means it was not, and a referee should compare it with the reference given.

For a finite sequence of nonnegative weights (a_n) , a set of primes $\mathcal{P}$ and $z \geq 2$, put $P(z) = \prod_{\ell \in \mathcal{P}, \ell < z} \ell$, $S(\mathcal{A}, \mathcal{P}, z) = \sum_{(n, P(z))=1} a_n$ and $A_d = \sum_{d|n} a_n$. Suppose that for squarefree $d \mid P(z)$

$$A_d = g(d)X + r_d, \quad (6)$$

with g multiplicative, $0 \leq g(\ell) < 1$, and put $V(z) = \prod_{\ell \in \mathcal{P}, \ell < z} (1 - g(\ell))$. The *dimension condition* (Ω_κ) is

$$\prod_{\substack{w \leq \ell < z \\ \ell \in \mathcal{P}}} (1 - g(\ell))^{-1} \leq \left(\frac{\log z}{\log w} \right)^\kappa \left(1 + \frac{K}{\log w} \right) \quad (2 \leq w < z).$$

Cited theorem 5.1 (arithmetic large sieve). Let S be a set of integers in an interval of length X which, for every prime ℓ in a finite set $\mathcal{P}$, avoids $\nu(\ell) < \ell$ residue classes modulo ℓ . Then for $Q \geq 1$, $|S| \leq (X + Q^2)/H(Q)$ with $H(Q)$ as in the proof of Lemma 4.1.

Montgomery [10], with the constant $X + Q^2$ from the Montgomery–Vaughan form of the analytic large sieve [11]; see also [4, §9.4]. *From memory*; it is the standard form and is used in the same way in [15, Lemma 12.1].

Cited theorem 5.2 (Mertens in progressions). For fixed $a \geq 1$ and $\gcd(c, a) = 1$, $\sum_{\ell \leq z, \ell \equiv c(a)} 1/\ell = \varphi(a)^{-1} \log \log z + M(a, c) + O_a(1/\log z)$. Consequently, if $\mathcal{P}$ is a union of k reduced classes modulo a , then

$$\prod_{\ell < w, \ell \in \mathcal{P}} (1 - 1/\ell)^{-1} = c_{\mathcal{P}} (\log w)^{k/\varphi(a)} (1 + O(1/\log w))$$

with $c_{\mathcal{P}} > 0$.

Standard consequence of the prime number theorem in arithmetic progressions; e.g. [12, §4.2].

Cited theorem 5.3 (Siegel–Walfisz). For every $B > 0$ there is $c_B > 0$ such that $\pi(t; a, c) = \text{li}(t)/\varphi(a) + O_B(t \exp(-c_B \sqrt{\log t}))$ uniformly for $a \leq (\log t)^B$ and $\gcd(c, a) = 1$.

[12, Cor. 11.21]. Standard; *from memory* as to the numbering. The constants are ineffective.

Cited theorem 5.4 (Bombieri–Vinogradov). For every $A > 0$ there is $B = B(A)$ such that $\sum_{k \leq x^{1/2}(\log x)^{-B}} \max_{\gcd(b, k)=1} |\pi(x; k, b) - \text{li}(x)/\varphi(k)| \ll_A x(\log x)^{-A}$.

Standard (e.g. [8, Thm. 17.1]); the version with $\max_{y \leq x}$ is not needed.

Definition 5.5 (Elliott–Halberstam conjecture, EH). For every $\eta, A > 0$: $\sum_{k \leq x^{1-\eta}} \max_{\gcd(b, k)=1} |\pi(x; k, b) - \text{li}(x)/\varphi(k)| \ll_{\eta, A} x(\log x)^{-A}$.

Cited theorem 5.6 (semi-linear sieve, lower bound). Assume (6) and $(\Omega_{1/2})$. Let $D = z^s$ with $1 \leq s \leq 2$. Then

$$S(\mathcal{A}, \mathcal{P}, z) \geq XV(z)(f(s) + o(1)) - \sum_{d < D, d|P(z)} |r_d|,$$

$$f(s) = \left(\frac{e^\gamma}{\pi s}\right)^{1/2} \int_1^s \frac{dt}{\sqrt{t(t-1)}},$$

where $o(1) \rightarrow 0$ as $D \rightarrow \infty$, uniformly for $s \in [1, 2]$, depending only on K .

Iwaniec [7]; in the β -sieve form [4, Thm. 11.13] with $\kappa = 1/2$, $\beta = 1$. *Partly checked:* the archived paper of Teräväinen [22, (6.4)] applies [4, Thm. 11.13] with exactly this f and $s \in [1, 2]$ (with error $O((\log N)^{-0.1})$ and the remainder weighted by the lower-bound sieve weights, which are bounded by 1). The general hypotheses as stated above are the standard ones and were *not* compared with [4] or [7]. We use only that, for $1 < s \leq 2$,

$$f(s) \geq c_f(s-1)^{1/2}, \quad c_f = (e^\gamma/\pi)^{1/2}, \quad (7)$$

which follows from $t(t-1) \leq 2(t-1)$ on $[1, 2]$, whence

$$\int_1^s \frac{dt}{\sqrt{t(t-1)}} \geq \int_1^s \frac{dt}{\sqrt{2(t-1)}} = \sqrt{2}(s-1)^{1/2},$$

and from $(e^\gamma/(\pi s))^{1/2} \geq (e^\gamma/(2\pi))^{1/2}$.

Cited theorem 5.7 (linear sieve, lower bound). Assume (6) and (Ω_1) . Let $D = z^s$ with $2 \leq s \leq 3$. Then

$$S(\mathcal{A}, \mathcal{P}, z) \geq XV(z)(f_1(s) + o(1)) - \sum_{d < D, d|P(z)} |r_d|,$$

$$f_1(s) = \frac{2e^\gamma \log(s-1)}{s},$$

with $o(1) \rightarrow 0$ as $D \rightarrow \infty$, uniformly in $s \in [2, 3]$, depending only on K .

Jurkat–Richert; [6, Thm. 8.4], [4, Thm. 11.13] with $\kappa = 1$, $\beta = 2$. *From memory* (the archived source [22, (6.6)] quotes only the upper linear bound). We use only $\log(1+u) \geq u/2$ on $[0, 1]$, whence

$$f_1(s) \geq \frac{e^\gamma}{3}(s-2) \quad (2 < s \leq 3). \quad (8)$$

Cited theorem 5.8 (Fuchs–Hsu–Rickards–Schindler–Stange). Let $f = ax^2 + bxy + cy^2$ be a primitive positive definite integral form of discriminant D with $\gcd(a, 2D) = 1$; let $A \neq 0$, $B \geq 1$ be integers with $\gcd(A, B) = 1$, and assume $2 \mid AB$ or $D \not\equiv 5 \pmod{8}$. Let $m \geq 1$, $\gcd(l, m) = 1$, $\gcd(m, 2DB) = 1$, $\gcd(l - A, m) = 1$. Then the number of primes $p \leq N$, $p \equiv l \pmod{m}$, primitively represented by $Bf(x, y) + A$ is $\gg N/(\log N)^{3/2}$.

[5, Thm. 1.1(2)]. *Checked* against the archived preprint (statement only; the proof was not checked). Used only in Remark 6.3 as a cross-check.

The proofs below use the sieve theorems only through (7) and (8); all upper-bound sieving is done with Lemma 4.1, which is proved above from Cited Theorem 5.1.

6. ONE WINDOW: $a_{\min}(p) \geq 7$ FOR $\gg x/(\log x)^{3/2}$ HARD PRIMES

Theorem 6.1 (PROVED modulo Cited Theorems 5.1, 5.2, 5.4, 5.6).

$$N_3(x) := \#\{p \leq x : p \equiv 1 \pmod{840},$$

$$(p+3)/4 \text{ has no prime factor } \equiv 2 \pmod{3}\} \gg \frac{x}{(\log x)^{3/2}}.$$

Every prime counted by $N_3(x)$ is Mordell-hard and has $a_{\min}(p) \geq 7$.

The second sentence holds because 1 is one of Mordell’s classes and by Lemma 2.6. The proof sifts the shifted primes $(p+3)/4$ by the 3-bad primes up to $z = x^{1/2-\varepsilon}$ with the semi-linear sieve, whose sifting limit $\beta = 1$ allows this at the Bombieri–Vinogradov level. At most two large bad primes then survive, the parity lemma removes the case of one, and the case of two is small.

Proof. Fix $0 < \varepsilon \leq 1/20$, to be chosen at the end as an absolute constant; $x \geq x_0(\varepsilon)$ is large. Let $B = B(2)$ be as in Cited Theorem 5.4, and put

$$z = x^{1/2-\varepsilon}, \quad D = \frac{x^{1/2}(\log x)^{-B}}{840}, \quad s = \frac{\log D}{\log z}.$$

Then $s \rightarrow 1/(1-2\varepsilon)$, so $1+\varepsilon \leq s \leq 2$ for large x .

Sieve data. Let $\mathcal{A}$ be the sequence $n_p = (p+3)/4$, $p \leq x$, $p \equiv 1 \pmod{840}$, and $\mathcal{P} = \{\ell : \ell \equiv 2 \pmod{3}\}$, the 3-bad primes. From $p \equiv 1 \pmod{840}$ we get $n_p \equiv 1 \pmod{210}$, so $2, 3, 5, 7 \nmid n_p$; put $g(2) = g(5) = 0$ (then $A_d = 0$ and $r_d = 0$ when $2 \mid d$ or $5 \mid d$). For squarefree $d \mid P(z)$ with $\gcd(d, 10) = 1$ we have $\gcd(d, 840) = 1$ and $d \mid n_p \iff p \equiv -3 \pmod{d}$; together with

$p \equiv 1 \pmod{840}$ this is one reduced class $b_d \pmod{840d}$. So (6) holds with

$$X = \frac{\text{li}(x)}{\varphi(840)} = \frac{\text{li}(x)}{192}, \quad g(d) = \frac{1}{\varphi(d)}, \quad r_d = \pi(x; 840d, b_d) - \frac{\text{li}(x)}{\varphi(840d)}.$$

For $\ell \in \mathcal{P}$, $\ell \geq 11$, one has $1 - g(\ell) = (1 - 1/\ell)(1 - (\ell - 1)^{-2})$, so Cited Theorem 5.2 (modulus 3) gives $(\Omega_{1/2})$ with an absolute K , and $V(z) \geq c_V(\log x)^{-1/2}$ with an absolute $c_V > 0$. The moduli $840d$, $d < D$, are distinct and at most $x^{1/2}(\log x)^{-B}$, so Cited Theorem 5.4 gives $\sum_{d < D} |r_d| \ll x(\log x)^{-2}$.

Step 1 (lower bound). By Cited Theorem 5.6 and (7), $f(s) \geq c_f \varepsilon^{1/2}$ and

$$\begin{aligned} S(\mathcal{A}, \mathcal{P}, z) &\geq \frac{\text{li}(x)}{192} c_V(\log x)^{-1/2} (c_f \varepsilon^{1/2} - o(1)) - O\left(\frac{x}{(\log x)^2}\right) \\ &\geq c_1 \varepsilon^{1/2} \frac{x}{(\log x)^{3/2}} \end{aligned}$$

for $x \geq x_0(\varepsilon)$, where $c_1 > 0$ is absolute.

Step 2 (survivors). Let $n = n_p$ be counted by $S(\mathcal{A}, \mathcal{P}, z)$ and $k = \Omega_3^-(n)$. All 3-bad prime factors of n are $\geq z$, so $z^k \leq n < x$ and $k < 1/(1/2 - \varepsilon) < 3$. Since $(\frac{p}{3}) = 1$, Lemma 2.5 says that k is even; so $k \in \{0, 2\}$. If $k = 0$ then p is counted by $N_3(x)$. If $k = 2$ then $n = mr_1 r_2$ with 3-bad primes $z \leq r_1 \leq r_2$ and m composed of primes $\equiv 1 \pmod{3}$. Hence

$$N_3(x) \geq S(\mathcal{A}, \mathcal{P}, z) - T_1 - T_2,$$

where T_1 counts the case $r_1 = r_2$ and T_2 the case $r_1 < r_2$.

Step 3. At most $x/(4r^2) + 1$ values n_p are divisible by r^2 , and only $r \leq \sqrt{x}$ occur, so $T_1 \leq \sum_{z \leq r \leq \sqrt{x}} (x/r^2 + 1) \leq 2x/z + \sqrt{x} \leq 3x^{1/2+\varepsilon}$.

Step 4. In T_2 we have $p = 4mr_1 r_2 - 3$. Fix m and r_1 , and put $a = 4mr_1$, $y = (x + 3)/a$. Then r_2 is a prime with $r_1 < r_2 \leq y$ and $ar_2 - 3 = p$ prime. Consequently $r_1 < \sqrt{x}$, $m \leq (x + 3)/(4z^2) \leq x^{2\varepsilon}$, and $y > z$, so $\log y \geq (\log x)/3$. Note $2 \mid a$ and $3 \nmid a$. Apply Lemma 4.1 to the integers $t \leq y$ with the primes $5 \leq \ell \leq \xi = y^{1/10}$, $\ell \nmid a$, and $\Omega_\ell = \{0, 3a^{-1}\}$ (two classes as $\ell \neq 3$). Every $r_2 > \xi$ counted avoids them, $\Lambda \leq 2 \log \xi + O(1) \leq (\log y)/4$, and

$$\prod_{\substack{5 \leq \ell \leq \xi \\ \ell \nmid a}} \left(1 - \frac{2}{\ell}\right) \leq C \left(\frac{a}{\varphi(a)}\right)^2 \prod_{5 \leq \ell \leq \xi} \left(1 - \frac{2}{\ell}\right) \leq C \left(\frac{a}{\varphi(a)}\right)^2 (\log y)^{-2},$$

using $(1 - 2/\ell)^{-1} = (1 - 1/\ell)^{-2}(1 + O(\ell^{-2}))$ for the removed primes $\ell \mid a$; here and below $C, C', \dots$ are absolute constants. Adding the at most ξ primes $r_2 \leq \xi$, the number of admissible r_2 is $\leq C'(a/\varphi(a))^2 y/(\log y)^2$, with C' absolute. Here m is odd and coprime to r_1 , so $a/\varphi(a) \leq 4m/\varphi(m)$, and

$$T_2 \leq C'' \frac{x}{(\log x)^2} \sum_{z \leq r_1 < \sqrt{x}} \frac{1}{r_1} \sum_{\substack{m \leq x^{2\varepsilon} \\ \ell \mid m \Rightarrow \ell \equiv 1 \pmod{3}}} \frac{(m/\varphi(m))^2}{m}.$$

By Mertens, $\sum_{z \leq r_1 < \sqrt{x}} 1/r_1 = \log \frac{1/2}{1/2-\varepsilon} + o(1) \leq 3\varepsilon$. The m -sum is at most $\prod_{\ell \leq x^{2\varepsilon}, \ell \equiv 1 \pmod{3}} (1 + \ell^2/(\ell-1)^3)$, and $(1 - 1/\ell)(1 + \ell^2/(\ell-1)^3) = 1 + (2\ell-1)/(\ell(\ell-1)^2) \leq 1 + 3/\ell^2$ for $\ell \geq 7$; so by Cited Theorem 5.2 it is $\leq C_2(2\varepsilon \log x)^{1/2}$. Therefore $T_2 \leq C_3 \varepsilon^{3/2} x/(\log x)^{3/2}$ with C_3 absolute.

Step 5. $N_3(x) \geq (c_1 \varepsilon^{1/2} - C_3 \varepsilon^{3/2}) x (\log x)^{-3/2} - 2x^{1/2+\varepsilon}$. With $\varepsilon = \min(1/20, c_1/(2C_3))$ the bracket is $\geq \frac{c_1}{2} \varepsilon^{1/2}$, and $N_3(x) \geq \frac{c_1}{4} \varepsilon^{1/2} x (\log x)^{-3/2}$ for $x \geq x_0$. $\square$

Remark 6.2. (i) The main term has size $\varepsilon^{1/2}$ and the two-prime configurations $\varepsilon^{3/2}$; the gain is the factor $\sum 1/r_1 \asymp \varepsilon$. This is the classical route of Iwaniec for primes $x^2 + y^2 + 1$ (compare [22, §6], which treats the analogous term with a linear upper sieve and a numerical inequality). (ii) The constant x_0 is ineffective, through Siegel's theorem in Bombieri–Vinogradov. (iii) The parity input is the congruence $(\frac{p}{3}) = 1$. Without it the statement is false: see Proposition 8.2.

Remark 6.3 (a second route). Cited Theorem 5.8 with $f = x^2 + xy + y^2$ ($D = -3$, $a = 1$), $B = 4$, $A = -3$, $m = 35$, $l = 1$ (all hypotheses hold: $2 \mid AB$, $\gcd(35, 24) = 1$, $\gcd(4, 35) = 1$) gives $\gg N/(\log N)^{3/2}$ primes $p = 4f(x, y) - 3 \equiv 1 \pmod{35}$ with $\gcd(x, y) = 1$. A primitive value $n = f(x, y)$ has no prime factor $r \equiv 2 \pmod{3}$: for odd such r , -3 is a non-residue and $(2x + y)^2 + 3y^2 \equiv 0$ forces $r \mid x, y$; and n is odd, since $x^2 + xy + y^2$ is even only if x, y are both even. As n is odd, $4n \equiv 4 \pmod{8}$ and $p = 4n - 3 \equiv 1 \pmod{8}$. Further $3 \nmid n$, since otherwise $3 \mid 4n - 3 = p$; and $n \equiv (x - y)^2 \pmod{3}$, so $n \equiv 1 \pmod{3}$ and $p \equiv 1 \pmod{3}$. Hence $p \equiv 1 \pmod{24}$, and with $p \equiv 1 \pmod{35}$, $p \equiv 1 \pmod{840}$. So [5, Thm. 1.1(2)] implies Theorem 6.1. Its proof runs Iwaniec's half-dimensional sieve; we cite it as a cross-check, not as an independent proof.

Remark 6.4 (numerics; EVIDENCE). A segmented sieve over $p = 840k + 1 < x$ gives $N_3(x)/(x(\log x)^{-3/2}) \in [0.0123, 0.0126]$ for $10^6 \leq x \leq 10^{11}$, with a slow drift (a slight rise from 10^6 to 10^7 , then a decrease; about -2.5% overall) (`scripts/window_w1.py`).

Corollary 6.5 (exact order for one window). *One has $\#\{p \leq x : p \equiv 1 \pmod{24}, a_{\min}(p) \geq 7\} \asymp x/(\log x)^{3/2}$; the upper bound is PROVED (Corollary 4.3 with $Z = 3$), the lower bound PROVED modulo the cited theorems of Theorem 6.1.*

7. TWO WINDOWS ON ELLIOTT–HALBERSTAM

Windows 3 and 7 together give a sieve problem of dimension one on the primes: for $\ell \nmid 840$ the forbidden classes of p are -3 if ℓ is 3-bad and -7 if ℓ is 7-bad, each of density $1/2$ among primes. The linear sieve has sifting limit $\beta = 2$, so sifting up to $z = x^{1/2-\varepsilon}$, which is what the parity argument needs, requires a level close to x .

Theorem 7.1 (CONDITIONAL on EH). *Assume EH (Definition 5.5). Let $N_{3,7}(x)$ be the number of primes $p \leq x$, $p \equiv 1 \pmod{840}$, such that $(p+3)/4$ has no prime factor $\equiv 2 \pmod{3}$ and $(p+7)/4$ has no prime factor r with $(\frac{r}{7}) = -1$. Then*

$$N_{3,7}(x) \gg \frac{x}{(\log x)^2},$$

and every prime counted is Mordell-hard with $a_{\min}(p) \geq 11$.

The second sentence is Lemma 2.4 at $a = 3$ and $a = 7$. Only a level $x^{1-\varepsilon_0}$ for one fixed small ε_0 is used, with a saving of $(\log x)^{-3}$. The proof also uses Cited Theorems 5.1, 5.2 and 5.7.

Proof. Write $n_3 = (p+3)/4$, $n_7 = (p+7)/4 = n_3 + 1$, so $\gcd(n_3, n_7) = 1$. Let $P_3 = \{\ell \equiv 2 \pmod{3}\}$ and $P_7 = \{\ell : (\frac{\ell}{7}) = -1\}$ be the 3-bad and 7-bad primes, and $\mathcal{P} = P_3 \cup P_7$. For $p \equiv 1 \pmod{840}$ we have $n_3 \equiv 1$, $n_7 \equiv 2 \pmod{210}$; so no prime $\ell \mid 840$ divides n_3 or n_7 except $2 \mid n_7$, and $2 \notin P_7$. Associate to p the integer

$$e_p = \prod_{\ell \in P_3} \ell^{v_\ell(n_3)} \prod_{\ell \in P_7} \ell^{v_\ell(n_7)}.$$

For squarefree d composed of primes of $\mathcal{P}$ with $\gcd(d, 840) = 1$, $d \mid e_p$ if and only if, for each $\ell \mid d$, $p \equiv -3$ (if $\ell \in P_3$) or $p \equiv -7$ (if $\ell \in P_7$) modulo ℓ , since $\gcd(n_3, n_7) = 1$. Let $\omega(\ell) = [\ell \in P_3] + [\ell \in P_7] \in \{1, 2\}$ for $\ell \in \mathcal{P}$, $\ell \nmid 840$, and $\omega(\ell) = 0$ for $\ell \mid 840$; the classes $-3, -7$ are distinct modulo $\ell \nmid 840$. Then the sequence $\mathcal{A} = (e_p)_{p \leq x, p \equiv 1 \pmod{840}}$ satisfies (6) with

$$X = \frac{\text{li}(x)}{192}, \quad g(d) = \prod_{\ell \mid d} \frac{\omega(\ell)}{\ell - 1}, \quad |r_d| \leq \tau(d) \max_{\gcd(b, 840d)=1} |E(x; 840d, b)|,$$

where $E(x; k, b) = \pi(x; k, b) - \text{li}(x)/\varphi(k)$ and τ is the divisor function. Sifting $\mathcal{A}$ by $\mathcal{P}$ up to z removes exactly the p for which n_3 has a 3-bad or n_7 a 7-bad prime factor below z .

Dimension and remainder. By Cited Theorem 5.2 for the moduli 3 and 7, $\sum_{w \leq \ell < z} \omega(\ell)/\ell = (\frac{1}{2} + \frac{1}{2}) \log \frac{\log z}{\log w} + O(1/\log w)$, and $-\log(1 - g(\ell)) = \omega(\ell)/\ell + O(\ell^{-2})$; so (Ω_1) holds with an absolute K , and $V(z) \geq c_V/\log x$. Let $\varepsilon \leq 1/20$, $z = x^{1/2-\varepsilon}$, $D = x^{1-\varepsilon}/840$. Using $\max_b |E(x; k, b)| \ll x/\varphi(k)$ for $k \leq x$, Cauchy-Schwarz and EH with $A = 10$,

$$\begin{aligned} \sum_{d < D} \mu^2(d) |r_d| &\leq \left(\sum_{d < D} \tau(d)^2 \frac{x}{\varphi(840d)} \right)^{1/2} \left(\sum_{k \leq x^{1-\varepsilon}} \max_b |E(x; k, b)| \right)^{1/2} \\ &\ll_\varepsilon (x(\log x)^4 \cdot x(\log x)^{-10})^{1/2} = \frac{x}{(\log x)^3}. \end{aligned}$$

Lower bound. Here $s = \log D/\log z \rightarrow (1-\varepsilon)/(1/2-\varepsilon) = 2 + 2\varepsilon/(1-2\varepsilon)$, so $2+\varepsilon \leq s \leq 3$ for large x , and (8) gives $f_1(s) \geq (e^\gamma/3)\varepsilon$. Cited Theorem 5.7 yields $S(\mathcal{A}, \mathcal{P}, z) \geq c_1 \varepsilon x/(\log x)^2$ for $x \geq x_0(\varepsilon)$, with $c_1 > 0$ absolute.

Survivors. For a sifted p and $q \in \{3, 7\}$, every q -bad prime factor of n_q is $\geq z$, so there are at most two (counted with multiplicity), and their number is even by Lemma 2.5, since $(\frac{p}{3}) = (\frac{p}{7}) = 1$. Hence

$$N_{3,7}(x) \geq S(\mathcal{A}, \mathcal{P}, z) - T_1 - T^{(3)} - T^{(7)},$$

where $T_1 \leq 2 \sum_{z \leq r \leq \sqrt{x}} (x/r^2 + 1) \leq 6x^{1/2+\varepsilon}$ counts sifted p with $r^2 \mid n_3 n_7$ for some $r \geq z$, and $T^{(q)}$ counts sifted p with $n_q = mr_1 r_2$, $z \leq r_1 < r_2$ both q -bad, m composed of q -good primes.

The terms $T^{(q)}$. Let q' be the other window. Fix m and r_1 and put $a = 4mr_1$, $y = (x + q)/a$. Then $r_2 \leq y$ is prime, $p = ar_2 - q$ is prime, and since $4n_{q'} = ar_2 + q' - q$, the integer $ar_2 + q' - q$ has no q' -bad prime factor $\ell < z$. As in Theorem 6.1, $r_1 < \sqrt{x}$, $m \leq x^{2\varepsilon}$ and $\log y \geq (\log x)/3$. Apply Lemma 4.1 to the integers $t \leq y$, the primes $11 \leq \ell \leq \xi = y^{1/20}$ with $\ell \nmid a$, and

$$\Omega_\ell = \{0, qa^{-1}\} \cup \{(q - q')a^{-1} : \text{if } \ell \text{ is } q'\text{-bad}\}.$$

These classes are distinct since $q, q', q - q' = \pm 4$ are nonzero modulo $\ell \geq 11$; $\nu(\ell) \leq 3$; every counted $r_2 > \xi$ avoids them (note $\xi \leq x^{1/20} < z$); and $\Lambda \leq 3 \log \xi + O(1) \leq (\log y)/4$. By Cited Theorem 5.2, $\sum_{\ell \leq \xi} \nu(\ell)/\ell = \frac{5}{2} \log \log \xi + O(1)$ when $\ell \mid a$ is not excluded, and excluding $\ell \mid a$ costs at most a factor $C(a/\varphi(a))^3$, because $(1 - 3/\ell)^{-1} \leq (1 - 1/\ell)^{-3}(1 + 6/\ell^2)$ for $\ell \geq 11$. So the number of admissible r_2 is $\leq C(a/\varphi(a))^3 y (\log y)^{-5/2}$. With $a/\varphi(a) \leq 4m/\varphi(m)$,

$$T^{(q)} \leq C' \frac{x}{(\log x)^{5/2}} \sum_{z \leq r_1 < \sqrt{x}} \frac{1}{r_1} \sum_{\substack{m \leq x^{2\varepsilon} \\ m \text{ } q\text{-good}}} \frac{(m/\varphi(m))^3}{m} \leq C'' \varepsilon^{3/2} \frac{x}{(\log x)^2},$$

by $\sum 1/r_1 \leq 3\varepsilon$ and $\sum_m (m/\varphi(m))^3/m \leq \prod_{\ell \leq x^{2\varepsilon}, \ell \text{ } q\text{-good}} (1 + \ell^3/(\ell - 1)^4) \leq C(2\varepsilon \log x)^{1/2}$ (the q -good primes have density $1/2$, and $(1 - 1/\ell)(1 + \ell^3/(\ell - 1)^4) = 1 + O(\ell^{-2})$).

Conclusion. $N_{3,7}(x) \geq (c_1 \varepsilon - 2C'' \varepsilon^{3/2})x(\log x)^{-2} - O(x^{1/2+\varepsilon})$, which is $\gg x/(\log x)^2$ once ε is a small absolute constant. $\square$

The essential point beyond Theorem 6.1 is that the other window's half-dimensional condition is kept inside the upper bound for $T^{(q)}$. Without it, $T^{(q)}$ would be of order $\varepsilon^{3/2}x/(\log x)^{3/2}$ and would swamp the main term.

Corollary 7.2 (exact order for two windows). *One has $\#\{p \leq x : p \equiv 1 \pmod{24}, a_{\min}(p) \geq 11\} \ll x/(\log x)^2$ (PROVED, Corollary 4.3 with $Z = 7$), and on EH it is $\asymp x/(\log x)^2$ (Theorem 7.1).*

Remark 7.3 (precedent). The shape of Theorem 7.1 is that of Friedlander and Iwaniec's hyperbolic prime number theorem [3], which counts, with multiplicity, the representations $p = x_1^2 + x_2^2 + x_3^2 + x_4^2$ with $x_1 x_4 - x_2 x_3 = 1$, so that $p - 2$ and $p + 2$ are both sums of two squares — two half-dimensional absence conditions on shifted primes — and whose lower bound assumes a

level of distribution close to 1. The final step is the same (removal of two-prime configurations using parity), but the sieve differs: [3] sift one condition with the semi-linear sieve and carry the other as a representation-number weight, while Theorem 7.1 sifts both absence conditions at once with the linear sieve at level $x^{1-\varepsilon}$. We claim no novelty of technique. The lower bound of [3] assumes a level x^θ with $\theta < 1$ sufficiently close to 1 ([3, Thm. 2 and Assumption $A(\theta)$]); Sedunova [21] (2026) recalls that it still rests on such a hypothesis and proves unconditional results for square-free distances instead. An unconditional result of the type “one complete absence and one almost-prime condition” is Nath–Xie [14, Thm. 1.1]: $\gg x/(\log x)^{5/2}$ primes $p = m^2 + n^2 + 1 \leq x$ with $\Omega(p+2) \leq 9$.

8. THE PARITY BARRIER

8.1. The two-window problem as a quadric.

Lemma 8.1 (PROVED). *Let p be Mordell-hard and $n = (p+3)/4$, so that $(p+7)/4 = n+1$. Windows 3 and 7 are both clean (no 3-bad prime factor of n , no 7-bad prime factor of $n+1$) if and only if $n = a^2 + ab + b^2$ and $n+1 = c^2 + cd + 2d^2$ with $\gcd(a, b) = \gcd(c, d) = 1$. Thus such p correspond to integral points of the quaternary quadric $c^2 + cd + 2d^2 - (a^2 + ab + b^2) = 1$ with $4(a^2 + ab + b^2) - 3$ prime.*

Proof. The forms are the norm forms $N(a - b\omega)$ of $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, and $N(c + d\theta)$ of $\mathbb{Z}[\theta]$, $\theta = (1 + \sqrt{-7})/2$; both rings are principal, and $a - b\omega$ (resp. $c + d\theta$) is not divisible by any rational integer > 1 iff $\gcd(a, b) = 1$ (resp. $\gcd(c, d) = 1$). For m coprime to the discriminant, m is the norm of such a primitive element iff no inert prime divides m : an inert $r \mid N(\alpha) = \alpha\bar{\alpha}$ divides α ; conversely, if every $r \mid m$ splits as $r = \pi_r \bar{\pi}_r$, then $\alpha = \prod \pi_r^{v_r(m)}$ is primitive of norm m . The inert primes are $r \equiv 2 \pmod{3}$ (including 2) for $\mathbb{Q}(\sqrt{-3})$, and the r with $(\frac{r}{7}) = -1$ for $\mathbb{Q}(\sqrt{-7})$ (2 splits as $-7 \equiv 1 \pmod{8}$, and $(\frac{-7}{r}) = (\frac{r}{7})$ for odd r). These are the 3-bad and 7-bad primes. Finally $3 \nmid n$ and $7 \nmid n+1$ since $p \equiv 1 \pmod{3}$ and $(\frac{p}{7}) = 1$. $\square$

In [3] the corresponding object is $p = x_1^2 + x_2^2 + x_3^2 + x_4^2$ on the quadric $x_1x_4 - x_2x_3 = 1$, i.e. $p \mp 2$ both sums of two squares.

8.2. Parity is necessary.

Proposition 8.2 (PROVED). *Let x be large and*

$$\mathcal{A}^\pm = \{p \leq x : p \equiv 1 \pmod{8}, p \equiv 1 \pmod{35}, \left(\frac{p}{3}\right) = \pm 1\},$$

and consider the sequences $n_3 = (p+3)/4$, $p \in \mathcal{A}^\pm$, sifted by $P_3 = \{\ell \equiv 2 \pmod{3}\}$.

(1) *(Same sieve data.) For every squarefree d composed of primes of P_3 ,*

$$\#\{p \in \mathcal{A}^\pm : d \mid n_3\} = g(d)X + r_d^\pm, \quad X = \frac{\text{li}(x)}{192},$$

$$g(d) = \begin{cases} 1/\varphi(d), & \gcd(d, 10) = 1, \\ 0, & \text{otherwise,} \end{cases}$$

with the same X and g for both signs, and for every A there is B with $\sum_{d \leq x^{1/2}(\log x)^{-B}} |r_d^\pm| \ll_A x(\log x)^{-A}$ (the moduli $840d$ are $\leq x^{1/2}(\log x)^{-B+1}$, so take $B = B'(A) + 1$ with B' from Cited Theorem 5.4). On EH the same holds for $d \leq x^{1-\eta}$, any fixed $\eta > 0$.

- (2) ($\mathcal{A}^-$.) For every $p \in \mathcal{A}^-$, n_3 has a prime factor $\equiv 2 \pmod{3}$, so window 3 succeeds and $S(\mathcal{A}^-, P_3, z) = 0$ for $z > (x+3)/4$.
- (3) ($\mathcal{A}^+$.) Window 3 fails for $\gg x/(\log x)^{3/2}$ primes $p \in \mathcal{A}^+$ (modulo the cited theorems of Theorem 6.1).

Proof. (1) As $p \equiv 1 \pmod{8}$, n_3 is odd, and $p \equiv 1 \pmod{5}$ gives $n_3 \equiv 1 \pmod{5}$; so $A_d^\pm = 0$ if $2 \mid d$ or $5 \mid d$. Otherwise $\gcd(d, 840) = 1$ (as $3, 7 \notin P_3$), and the conditions $p \equiv 1 \pmod{8}$, $p \equiv 1 \pmod{35}$, $p \equiv \pm 1 \pmod{3}$, $p \equiv -3 \pmod{d}$ define one reduced class $b_d^\pm$ modulo $840d$. So $A_d^\pm = \pi(x; 840d, b_d^\pm)$ and $r_d^\pm = \pi(x; 840d, b_d^\pm) - \text{li}(x)/\varphi(840d)$, and Cited Theorem 5.4 (resp. EH) bounds the remainders. (2) $4n_3 = p + 3 \equiv p \pmod{3}$ gives $n_3 \equiv p \equiv 2 \pmod{3}$, so some prime factor of n_3 is $\equiv 2 \pmod{3}$, and window 3 succeeds by Lemma 2.6's argument ($u = r$, $v = 1$). (3) $\mathcal{A}^+$ contains the primes $p \equiv 1 \pmod{840}$ counted by Theorem 6.1. $\square$

Consequently, any lower-bound method for $S(\mathcal{A}, P_3, z)$ whose input is only X , g , the level D and a bound for $\sum_{d < D} |r_d|$ (the axioms of [4, Ch. 11]) returns a non-positive lower bound for $z > (x+3)/4$ on the data of $\mathcal{A}^+$, at every level $D \leq x^{1/2}(\log x)^{-B}$ unconditionally and every $D \leq x^{1-\eta}$ on EH. A proof that window 3 fails for some p must use $(\frac{p}{3}) = +1$, i.e. Lemma 2.5. This is Selberg's parity example, with the Liouville function of the bad part realised as the Dirichlet character $(\frac{p}{3})$ of the prime itself. Because it is a character of p , the barrier is removed by a congruence condition, which is exactly what Step 2 of Theorem 6.1 does.

Remark 8.3 (two windows jointly). For windows 3 and 7 jointly, the four sign classes $((\frac{p}{3}), (\frac{p}{7}))$ do *not* all have the same joint sieve data: 3 is 7-bad, and $3 \mid n_7 = (p+7)/4$ for every $p \equiv 2 \pmod{3}$ and for no $p \equiv 1 \pmod{3}$. The barrier holds for the pair of classes

$$\mathcal{B}^\pm = \{p \leq x : p \equiv 1 \pmod{8}, p \equiv 1 \pmod{5}, p \equiv 1 \pmod{3}, \left(\frac{p}{7}\right) = \pm 1\}.$$

For coprime squarefree d_1 (composed of 3-bad primes) and d_2 (composed of 7-bad primes), $\#\{p \in \mathcal{B}^\pm : d_1 \mid n_3, d_2 \mid n_7\}$ vanishes if $\gcd(d_1 d_2, 30) > 1$ (as n_3 is odd, $3 \nmid n_7$ and $5 \nmid n_3 n_7$), and otherwise is a sum of three prime counts in reduced classes modulo $840d_1 d_2$, with main term $3 \text{li}(x)/\varphi(840d_1 d_2)$ for both signs. (For $\gcd(d_1, d_2) > 1$ both counts vanish since $\gcd(n_3, n_7) = 1$.) Yet no $p \in \mathcal{B}^-$ has window 7 clean, since $(-1)^{\Omega_7^-(n_7)} = (\frac{p}{7}) = -1$, while on EH Theorem 7.1 gives $\gg x/(\log x)^2$ primes of $\mathcal{B}^+$ with both windows clean. So the parity of window 7 is also necessary input.

Remark 8.4 (why two windows are harder). Two windows form a dimension-one problem. At the Bombieri–Vinogradov level $x^{1/2}$ the linear sieve reaches only $z < x^{1/4}$, so each window can carry two or four large bad primes, and the configurations with two bad primes are no longer of lower order. Lemma 2.5 still removes odd counts, but it does not distinguish 0 from 2 or 4. Whether Type I information at level $x^{1/2}$ together with the parity of both windows suffices is the question addressed, in a model only, in Remark 9.6.

9. HEURISTICS, DATA AND A MODEL-LEVEL OBSTRUCTION

Nothing in this section is a theorem about primes unless so labelled.

Remark 9.1 (random model; ASSESSMENT). By Lemmas 2.4 and 3.1, failure of a fixed window is a confinement of the prime factors of x_a to half of the classes, an event of probability $\asymp (\log p)^{-1/2}$ for a random integer of size p ; Theorem 4.2 with $|A| = 1$ is the corresponding upper bound on shifted primes. Treat the windows as independent, each failing with probability $g(p) = c(\log p)^{-1/2}$. Then $\Pr(a_{\min}(p) > Q) \approx g(p)^{(Q+1)/4}$, which gives (2) for fixed Q , and $\log \Pr(a_{\min}(p) > Q) = -(Q/8)(1 + o(1)) \log \log p$ for growing Q . This is summable over p once $Q \geq (8 + \delta) \log p / \log \log p$, so the model predicts $a_{\min}(p) \leq (8 + o(1)) \log p / \log \log p$ for all large p , while congruence-adversarial constructions (Remark 9.2) heuristically produce $p \leq N$ with $a_{\min}(p) \geq c \log N / \log \log N$. The model ignores the positive correlation of windows through the quadratic-residue pattern of p (by Lemma 2.3, every prime factor of x_a below the least non-residue modulo p is a -good), and it applies a fixed-window law to windows of growing size, which is not justified. Theorems A–C show that the model’s exponent is a true upper bound for each fixed Z , and that it is attained for $Z = 3$ (and for $Z = 7$ on EH). For growing Z the proved bounds (Theorem 4.4 and the remark after it) are far from the model.

Remark 9.2 (unboundedness of $a_{\min}$; CONDITIONAL on Dickson’s conjecture). Fix $K \geq 3$ and let Λ_K be the set of primes $\leq \max(K, 5)$. If $p \equiv 1 \pmod{24}$ is a quadratic residue modulo every $\ell \in \Lambda_K$ and, for every $a \equiv 3 \pmod{4}$ with $a \leq K$, $x_a = C_a r_a$ with r_a prime and all prime factors of C_a in Λ_K , then $a_{\min}(p) > K$ (PROVED): the primes of C_a are residues modulo p by reciprocity, $\left(\frac{r_a}{p}\right) = \left(\frac{4x_a}{p}\right) = \left(\frac{a}{p}\right) = \prod_{\ell|a} \left(\frac{p}{\ell}\right)^{v_\ell(a)} = 1$, and Lemma 2.4 applies. Fixing the residue of p modulo a suitable power of $\prod_{\ell \in \Lambda_K} \ell$ turns this into the simultaneous primality of $J(K) + 1$ linear forms, an admissible family [16, Prop. 8.4]; Dickson’s conjecture [1] then gives infinitely many such p , so $a_{\min}$ would be unbounded. The only unconditional result of this kind that we know of is $a_{\min}(p) \geq 7$ infinitely often (Theorem 6.1).

Remark 9.3 (data; EVIDENCE). A complete census of the 719781 primes $p \equiv 1 \pmod{24}$ below 10^8 (among them the 179468 Mordell-hard ones) gives $\max a_{\min}(p) = 107$, attained at $p = 8803369$, a quadratic residue modulo every prime ≤ 37 ; the ratio $a_{\min}(p) / \log p$ is below 6.7 throughout.

Random samples of $3 \cdot 10^4$ Mordell-hard primes near 10^{12} and 10^{18} have maxima 39 and 43, and samples near 10^{24} also stay below $10 \log p$. For $x = 10^6, 10^7, 10^8$ and each $Z \in \{3, 7, 11, 15, 19, 23\}$ the normalised tail $\#\{p < x : p \equiv 1 \pmod{24}, a_{\min}(p) > Z\} \cdot (\log x)^{1+J(Z)/2}/x$ is constant to within 9%, consistent with Corollary 4.3 being sharp in the exponent for every fixed Z (proved only for $Z = 3$, and for $Z = 7$ on EH). These data and the model motivate the following conjecture, which by Theorem 2.2, Mordell's identities and the numerical verification of the Erdős–Straus conjecture up to 10^{18} [9] would imply the Erdős–Straus conjecture.

Conjecture 9.4 ($X_{\text{win}}(10)$; CONJECTURE). *For every prime $p \equiv 1 \pmod{24}$ with $p > 10^{18}$, $a_{\min}(p) \leq 10 \log p$.*

9.1. Type I information plus parity in a discrete model. The following describes a finite computation in an explicit model. It is recorded because it indicates what kind of input a proof of unconditional $a_{\min}(p) \geq 11$ infinitely often might need, not as a result about primes.

The model. For a prime p let the *configuration* $C = (B_3, B_7)$ consist of the multisets of log-sizes $t = \log r / \log x$ of the 3-bad prime factors of n_3 and the 7-bad prime factors of n_7 . *Parity* means $|B_3|, |B_7|$ even (Lemma 2.5 for Mordell-hard p). For a pair of finite multisets $S = (S_3, S_7)$ let $\rho_\mu(S)$ be the expected number of embeddings of S into C under a law μ ; the *Type I data of level θ* are the $\rho_\mu(S)$ with $\Sigma S \leq \theta$ (sum of all entries), corresponding to the counts $\#\{p : d_1 \mid n_3, d_2 \mid n_7\}$ for bad squarefree d_1, d_2 with $d_1 d_2 \leq x^\theta$. A *fake* for μ at level θ is a nonnegative measure ν on parity configurations with $\rho_\nu(S) = \rho_\mu(S)$ for all $\Sigma S \leq \theta$. If a fake has $\nu(\emptyset, \emptyset) = 0$, then these data together with parity cannot force a configuration with both windows clean.

Remark 9.5 (two elementary fakes; MODEL, proved). (i) One window, $\theta < 1/2$: for $u, v \in (\theta, 1/2)$ the signed measure $-\emptyset + \{u, v\}$ has $\rho(S) = 0$ whenever $\Sigma S \leq \theta$, since every nonempty $S \subseteq \{u, v\}$ has $\Sigma S > \theta$. Hence for any true law μ the measure $\mu + \mu(\emptyset)(-\emptyset + \{u, v\})$ is a fake without clean mass. For $\theta \geq 1/2$ there is no such pair, since $u, v > \theta$ would force $u + v > 1$. (ii) Two windows: if U, V are disjoint nonempty configurations, each with an even number of points in each window and $\min U + \min V > \theta$, then $-\emptyset - [U \sqcup V] + [U] + [V]$ has $\rho(S) = 0$ for $\Sigma S \leq \theta$: for $S = \emptyset$ the terms cancel, for $\emptyset \neq S \subseteq U$ (or $\subseteq V$) the two terms containing S cancel, and an S meeting both U and V has $\Sigma S > \theta$. Adding c times this signed measure to μ keeps it nonnegative only if $c \leq \mu(U \sqcup V)$, so a fake without clean mass arises this way if $\mu(U \sqcup V) \geq \mu(\emptyset)$, or from a combination of several such blocks whose removable masses add up to $\mu(\emptyset)$.

Remark 9.6 (a certified fake at level $1/2$; MODEL, certified in floating point). Take the heuristic law in which the two windows are independent, bad primes have Mertens density $\frac{1}{2} dt/t$ on $[\varepsilon, 1]$ and the clean cofactor has weight $(1 - \Sigma t)^{-1/2}$, and discretise it with $\varepsilon = 0.1$ and 8 log-spaced bins.

In this discrete model a linear-programming search, followed by an exact re-solve on its support, finds a fake at level $\theta = 1/2$ with $\nu(\emptyset, \emptyset) = 0$: an 89×89 system over 12769 configurations, nonnegative solution, relative correlation residual $2.6 \cdot 10^{-15}$, independently re-solved at 60 digits. In the same model one window cannot be faked at $\theta = 1/2$: an exact rational dual certificate gives $\min \nu(\emptyset)/\mu(\emptyset) \geq 0.744$. For two windows the threshold level lies in $(0.5, 0.7]$ on this grid (certified dual bound ≥ 0.497 at $\theta = 0.7$).

Scope. This is a statement about a coarse discrete model only. The law is heuristic and not derived from primes; at $\varepsilon = 0.1$ its normalised one-window correlations vary over $[0.82, 1.87]$ and differ between the even and odd laws, whereas for primes Bombieri–Vinogradov and the fundamental lemma make them ≈ 1 independently of parity (Proposition 8.2); grid refinement did not remove this; the model contains only the data for bad d , not for good or mixed d ; and the fake places up to 10^5 times the true mass on some configurations, with no capacity constraint. Its relevance to actual sieve methods is EVIDENCE of the weakest kind. The indication is that Type I data of level $1/2$ plus parity do not decide two windows, and that a Chen-type switching input bounding large-prime configurations would be needed; no such statement is proved here. Computations: `scripts/window2_lp.py`, `window2_feas.py`, `window2_polish.py`, `window2_verify.py`; details in [19, §3].

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- [16] Companion note in this repository, `POINTWISE_SIZE.md`, §8 (the window frame); internally reviewed in `reviews/pointwise-size-step1-review.md`.
- [17] Companion note in this repository, `POINTWISE_WINDOW.md` (Theorems W1, W2); internally reviewed in `reviews/pointwise-window-review.md`.
- [18] Companion note in this repository, `POINTWISE_XWIN.md` (half-set lemma, stacking); internally reviewed in `reviews/pointwise-xwin-review.md`.
- [19] Companion note in this repository, `POINTWISE_WINDOW2.md` (quadric, Theorem P1, the discrete model); internally reviewed in `reviews/pointwise-window2-review.md`.
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